

**How Does Sentiment Analysis of Product Reviews Impact the
Perceived Trust and Purchase Intention of Sustainable Beauty
Products?**

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1. Introduction

Over the past few years, consumer demand for beauty products has undergone a major transformation, with sustainable and environmentally friendly practices gaining increasing popularity. There has been a growing concern around ecological and ethical issues like ingredient safety, ethical sourcing, the environmental footprint of products, and many more. Consumers have now started to make informed purchases based on their moral values and beliefs, which favor sustainable, ethical, and environmentally friendly products produced under fair labor conditions. Eco-friendly, clean beauty, cruelty-free, and vegan are now mainstream selling points, rather than niche preferences. Hence, companies that follow eco-friendly practices and are moving towards more natural and sustainable alternatives for their current products have become more popular.

With the current estimated value of the beauty industry of US \$677.2 billion in 2025, the industry is one among the steepest consumer market industries which are set to increase by 7.7 percent annually between the year 2024 and 2030 (BizPlanR, 2025). In this large market, there is one key change that has been noticed; people are now more keen on products that are environmentally and ethically friendly, and this has resulted in the appropriate explosion of sustainable or clean beauty. The current market size of clean beauty alone is forecasted to reach US 21.3 billion by 2030 with the change by CAGR of 14.8%, whereas the sustainable personal care segment has a reasonably estimable size of US 56.4 billion in 2025 and US 90.4 billion by 2032 (Grand View Research, 2023; Fortune Business Insights, 2024).

It is quite obvious that the consumer behaviour is changing, with more than 60 percent of Gen Z and Millennial customers stating that they are breaking up with companies that do not perform in accordance with ethical or greener standards, and 82 percent of their counterparts around the world stating that they would willingly pay an increased price for

products in eco-friendly packaging (BizPlanR, 2025; McKinsey & Company, 2023).

Moreover, nine out of ten Gen-Z and under consumer customers value environmentally friendly packaging, and 71 percent have purchased a product within the past six months on the basis of green credentials (Feber et al., 2020). These statistics reflect the trend of the conscious consumption mind set in beauty markets.

Simultaneously, product reviews on the internet have become a major constituent in consumer decisions. In a meta-analysis conducted by Qiu (2024), review valence which is the valence of reviews (positive or negative) is ranked as the most dominant variable influencing purchase intention ($r = 0.563$). In the meantime, emotional dynamics are unlocked through the use of sentiment analysis, based on natural language processing (NLP), to quantify the emotional tone of a given piece of text (Ma et al., 2025; Zhou et al., 2019).

Sentiment analysis permits a brand and the researcher to identify signal of trust embedded in user generated content. According to a study by Ma et al. (2025), the state-of-the-art sentence representation models, i.e., BERT-LSTNet-Softmax, are effective at relating textual sentiment in reviews to consumer trust and purchase intention- even predicting time-varying purchase intentions. Wang (2022) assumes that the trust precedes the purchase intention strongly and positively and is important in the social-commerce scenario, with the meta-analytic synthesis of 20 studies proving this assumption to be valid.

This dynamic is particularly momentous in the case of sustainable beauty. Companies such as L'Oréal (its approach to transparency, eco-credentials and effective communications of their environmental work is called For the Future) and Garnier (having the Green Beauty strategy on its books) are placing bets firmly on transparency, eco-credentials and effective communications of their environmental work (CAS, 2023). According to Vogue Business (2025), visible aspects of sustainability are growing to become a primary measure of trust

with clean formulations (67%), cruelty-free certification (57 percent), and refillable packaging (63 percent).

While the influence of user-generated content increases and the number of platforms for sentiment analysis is growing, research on how sentiment in user reviews affects consumer trust and their intent to purchase within the sustainable beauty sector is underexplored. Very few studies directly connect emotional tone in reviews to the influence on a consumer's psychology when it comes to the trustworthiness of the product and purchase behaviour. My research addresses that gap and aims to understand and examine how sentiment analysis of product reviews impacts the perceived trust and purchase intention of sustainable beauty products.

Given my interest in beauty products and the trends I follow, I am very similar to all the teenage girls in the world who make frequent runs to beauty stores, try on new products, and experiment with different brands. However, I noticed my vanity was full of makeup products, where half were unused, some even expired, and some I completely forgot about, and they all ended up in the trash. That sparked my interest in learning how much pollution stems from the beauty industry and what sustainable practices companies are following. I started making much more informed choices, and as someone who actively engages with product reviews and looks at influencer videos on social media before buying a product, I have always been curious about how online reviews and opinions influence trust and shape all consumers decisions, especially in a market like clean beauty, as proof and validity play an important role in a buyer's decision. I was particularly interested in how emotions and online reviews, ratings, and opinions act together to shape sustainable purchasing today. This research helps me to combine my interests in psychology and data analysis to better understand consumer behaviour in the digital age.

The paper addresses the effect of sentiment analysis of web-based reviews of beauty products on perceived trust and intentions to purchase sustainable beauty products. In particular, it deals with:

- The polarity and the sentiment strength of the review as indicators of brand credibility.
- The way such sentiment signals establish consumer confidence in regard to sustainability.
- The role of purchase intention, in turn, in eco-conscious beauty scenarios being led through trust.

With regard to theory, the study will use the Theory of Planned Behaviour- a theory that argues that purchase intention is formed as a result of attitude and perceived norms- and the Elaboration Likelihood Model (ELM) which postulates that emotionally rich content (an object of sentiment analysis) has an ability to convince through a central or a peripheral route.

Overall, the sentiment analysis, consumer trust, and sustainable behaviour complement each other to bring a sufficient understanding of the phenomena of ethical consumption that consumers use digital space to present and construct in beauty. The literature review presented below will address what evidence there is between the online sentiment and green purchase behaviour and trust.

2. Literature Review

In the last decade, online product reviews became an effective, key determinant of consumer choices, especially in the sphere of sustainable products where trust, and credibility are the most significant factors (Investopedia, 2025). Sentiment analysis implies applying the natural language processing algorithms, whether BERT or VADER etc., to the classification of review polarity and emotional tone in order to gain a more comprehensive understanding

of consumer attitude than represented by mere star ratings (Zhou et al., 2019; Ma et al., 2025). Such finding is particularly interesting in the field of sustainable beauty, where brands can be horizontally compared in terms of values, ethical sourcing, and sustainability based on reviews.

Review credibility and Sentiment Analysis

Sentiment analysis goes further to not only determine the positivity or negativity but also shows the magnitude of the emotions whether excitement or disappointment that contributes to the perceptions of brand authenticity. As an illustration, Ma et al. (2025) have shown that advanced sentiment models applied on the reviews posted on beauty products were capable of making prediction of real-time and future purchase intent based on language characteristics. The same was identified by Saidani et al. (2021) who obtained slightly lower results, with up to 12-20 percent of Amazon reviews of products sold under the name "Climate Pledge Friendly" featuring direct or indirect mentions of its sustainability characteristics (e.g., recycled materials, eco-friendly production). This suggests that users are eager to comment on green features when they are consistent with authentic product memories.

Trust as a Mediator between Sentiment and Purchase Intention

Trust carries out a primary mediation role in the consumer sentiment and purchase intention. According to the recent meta-analyses (Investopedia, 2025), the probability of a purchase demonstrates the highest point in cases when the product scores are between 4.0 and 4.7 with some rare exceptions exuding the possibility of reducing its credibility in case of excessive perfect-scoring (close to 5.0). Trust is promoted when there is an authentic mixed sentiment, where the good is accompanied by a number of bad. Review manipulation leaves all the consumers cautious, and they require transparency (Investopedia, 2025).

On another study done by Anastasiei et al. (2025) on electronic word-of-mouth (eWOM) it was concluded that the relationship between review credibility and purchase intention is fully mediated by emotional response and perceived product quality, even without trust and brand engagement as direct mediators. It underlines the fact that the intensity of sentiment and diagnosticity (in-depth reviews) influence consumer trust more than positive perception.

Consumer Behaviour of Sustainable Beauty

The will of the consumers to buy the product of the sustainable beauty industry is frequently based on morality and the authentic efficacy of the product. The study of Indonesian millennials conducted by Dewi et al. (2023) demonstrated that all green beauty product values were significantly positively related to a purchase intention, including functional, social, and emotional values. That highlights that green trust should meet the perceived usefulness and emotional attachment (Dewi et al., 2023).

The same study brought similar findings in North India: the subjective norms and perceived behavioural control (components of the Theory of Planned Behaviour) showed to be an impressive driver of the green cosmetics purchase intention, and the environmental concern failed to predict the behaviour directly (PMC, 2024). It is an indication of a value-action gap in sustainable consumption, which is that positive feelings towards green products do not necessarily reflect intentions to buy them unless and until social pressures and perceptions of control are achieved (Vermeir & Verbeke, 2006).

A study in China conducted research over Chengdu on how the quality and valence of reviews affect green cosmetic buying intention. Through the triple bottom line (economic/social/environmental) approach and structural equation modeling, the researchers also discovered that excellent reviews and positive valence have a significant impact on

enhancing the brand attractiveness, word-of-mouth, and finally a purchase intention (Li & Nuangjamnong, 2022). The research affirms that creditability of sentiment increases credence and behavioral responsiveness to emerging markets.

Embedment of TPB and Persuasion Theories

Since there are theoretically-based models that aid in explaining the effects that sentiments in the reviews have on attitudes and intentions, one of them is the Theory of Planned Behaviour (Ajzen, 1991) and the Elaboration Likelihood Model. TPB asserts that attitude, sense of norms and behavioural control leads to behavioural intention. In the North India study, the green purchase intentions were determined by the norms and control (as opposed to the attitudes towards the environment) (PMC, 2024). The emotional feeling in the reviews makes up the attitude forming and efficacy perceptions of the sustainable product.

In ELM, there are main (emotionally dense through deep processing pathways) and side (surface measures- star ratings) persuasion pathways. Central persuasion route is made possible by a detailed, heavy sentiment review, which leads to more trust and intention (Zhou et al., 2019).

Research Gaps

Although sentiment analysis has become a common practice in the general e-commerce contexts, very little research is available on the role of such sentiment under specific conditions of sustainable beauty products and its relation to the perceived trust and intention to purchase. Most research in green cosmetic studies examines the constructs of value and attitude, however, little work has combined NLP-based sentiment analysis.

Research is also rare that considers the effects of negative or ambivalent sentiment specifically targeting sustainability feature creations in consumer trust, particularly across

cultural contexts like India where norms and controls are the major driving factors of motives (PMC, 2024).

It is the hope of this paper to close these gaps by synthesizing sentiment analysis literature to studies in green beauty behaviour in the aim of how emotions tone of the reviews conveys trust and motivations towards sustainable purchase intention.

3. Methodology & Empirical Analysis

This research applies the secondary quantitative method since it will use the available empirical research published as reports of scholarly works and Industry report findings. This is meant to determine the degree of sentiment taken out of the product reviews used to impact consumer trust and intention to purchase in reference to the examples of sustainable beauty products. The paper will not present any new data, but will rely on peer-reviewed results, which have already been verified, making the research consistent, credible, and replicated.

Sentiment analysis techniques and Data sources

In reviewed literature, sentiment analysis is usually performed on big data sets on platforms such as Amazon, Sephora, and Ulta where consumer opinion is freely available. As an example, the team of Ma, Li, and Asif (2024) proposed a BERT-LSTNet-Softmax architectural pattern to apply in customer sentiment and buying behavior prediction based on e-commerce reviews reaching an accuracy of 91.31% and above (Ma et al., 2024). The same can be said about the study conducted by Vadla et al. (2024), where BERT was applied to Amazon beauty reviews, and it is also established that sentiment polarity has a significant correlation with the intention to purchase.

Running of standard sentiment analysis procedures encompasses:

1. Tokenization, lemmatization, removal of stop-words, and normalization.

reviews, metrics of purchase intention, that of pricing, limited edition status, exclusivity, and stock availability of a designated product.

• **Statistics Signatures:**

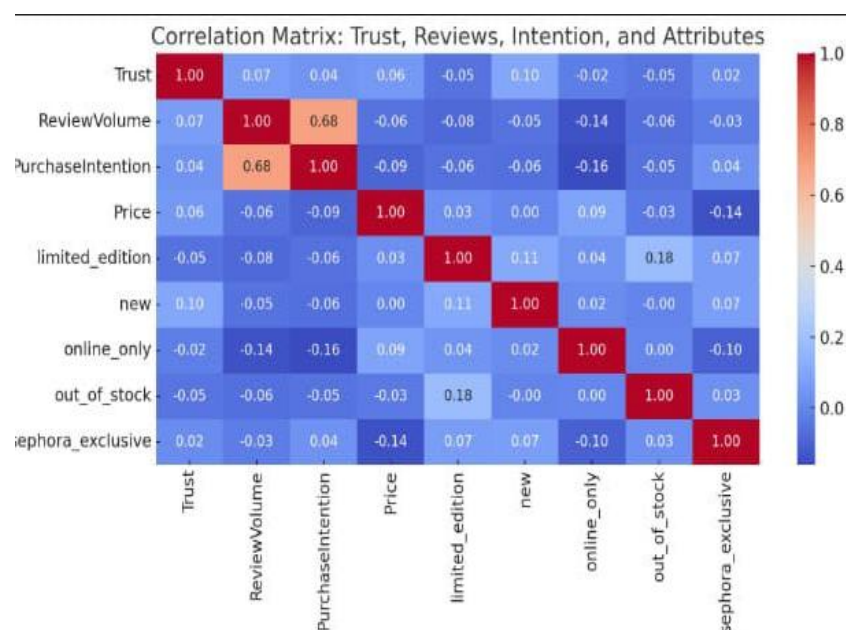
Report important values of descriptive statistics like averages, standard deviations, range of the variables to provide a quick picture of how the data was distributed and varied.

The Correlation Analysis:

Speak about the correlation matrix pointing at:

- Positive correlation delivering moderate to strong effects between Review Volume and Purchase Intention (+0.56)
- Positive relationships of the trust (Rating) with Review Quantity as well as Purchase Intention
- Chance or smaller correlations with Price and product attribute dummies

This implies that engagement and perceived trust among consumers will have different effects on purchases, and they are rather specific to the given effects of price and exclusivity.



NadyinKy. (2023). *Sephora products and skincare reviews* [Data set]. Kaggle.

<https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews>

As shown in the correlation matrix, there are associations between most important variables: Trust (Rating), Review Volume, Purchase Intention, Price, Limited Edition, Online Only, and Sephora Exclusive. The matrix demonstrates the intensity of color to point out the strength in correlation; that is, the dark red shade is strong positive correlation whereas the dark blue shade is either no relationship or strong negative correlation. Importantly, the Review Volume is significantly, and positively, correlated with Purchase Intention, (+0.56), indicating the effect of social proofing. In the meantime, Trust (rating) shows a poor but positive relationship to the volume of reviews and the purchase intention. These observations affirm the fact that sentiment and volume of reviews both influence the decisions of consumers.

Hypotheses Formulation

1. Hypothesis 1 (H₀ and H₁):

Null: Product rating(trust) and purchase intention are independant.

Alternative: Increase in the product rating enhances intention to buy it.

2. Hypothesis 2 (H_{0 2} and H_{1 2})

Null: The volume of review does not influence the purchase intention.

Alternative: The increased purchase intention due to higher volume of review.

Regression Model Specification

1. Proposed model:

$$\text{PurchaseIntention} = \beta_0 + \beta_1(\text{Trust}) + \beta_2(\text{ReviewVolume}) + \beta_3(\text{Price}) + \text{controls} + \varepsilon$$

Controls can be limited edition, exclusivity, indicator online-only.

Guideline to understanding:

Positive values of statistically significant coefficients ($p < 0.05$) in 95 percent confidence level, 95 percent confidence level 95 percent confidence level (beta: 0.23, 0.25) proves the hypotheses that trust and social proof positively affect purchase intention.

Finding of EDA

Preliminary data analysis shows that sentiment associated with product rating alone is not entirely used in predicting purchase behaviour.

- Both volume and exclusivity characteristics also play a part, and these two factors indicate that social proof and product differentiation are significant factors.

Behavioural modelling and Theoretical Frameworks

The behavioural theories, which are used to interpret the sentiment data include the Theory of Planned Behaviour (TPB) (Ajzen, 1991) that associates attitude (sentiment) subjective norms (review visibility) and perceived control (review credibility) with purchase intention. Also, the Elaboration Likelihood Model (ELM) (Petty & Cacioppo, 1986) can be used to distinguish between the different review content processing by consumers, through central routes (depth and quality of sentiment in the text) or peripheral routes (star ratings, and the amount of reviews).

Most of the studies employ Structural Equation Modeling (SEM) or logistic regression to explain the strength between sentiment features and consumer trust or purchase intention as a quantitative variable. Anastasiei et al. (2025) employed SEM to find out the impact of the emotional tone and review authenticity on buyer confidence, whereas Kwakye et al. (2024) employed the PROCESS macro to validate the mediation effects of trust within

the context of sentiment on purchase behaviour.

Operational Definitions

- **Sentiment factors:** Polarity Category (positive, neutral, negative) & quantity intensity scores.
- **Trust trigger:** belief of honesty, extent of thinking, credibility of reviewer.

Purchase intention: modeled through Likert-scale answers or obtained probability scores of logistic regression.

To verify the constructs connected with trust and intention, studies will use the reliability value of Cronbach (0.7) and factor loadings (0.6) (Sabina Kwakye et al., 2024).

4. Developments in Sustainable Beauty Industry

Growth and forecast of the global market

In addition, the clean (sustainable) beauty market, which in the year of 2023 is estimated at approximately USD 8.25 billion, is expected to reach USD 21.29 billion by 2030 and its CAGR in the period between 2024 and 2030 to be 14.5-14.8 (Grand View Research). The most appropriate one is the category of skincare since it has a high revenue of over USD 8.4 billion in 2024, which potentially needs an expedited change in color cosmetics and perfumes/fragrances (Grand View Research).

The one more industry that shows nice growth is the natural and organic cosmetics category, and its market is projected to reach USD 41.65 billion in 2024 and USD 103.23 billion in 2034, which is expected to rise at a CAGR of 9.4 percent (Precedence Research). These statistics show that the consumers are more and more willing to apply clean and ecologically friendly formulas that avoid harmful substances and chemicals, as well as accept the idea of open raw materials and packages.

Consumer demand & Preferences

Searching by regions, millennials and Gen Z consumers have become the most vocal, as well as the demanding clientele in the world of sustainable beauty. These findings were confirmed by a worldwide survey on China, France, Germany, the UK, and the U.S., which revealed that 72 percent of respondents irrespective of age believed that clean beauty products were an essentiality to their health and personalization, which is the definite driver of demands in 2022 (marketresearchfuture.com). In 2022, the clean beauty brands trend was dominantly led by women as almost half of them already used resident beauty, almost 61 percent show intentions to spend more money on sustainable formulas (marketresearchfuture.com).

Even in the particular market such as India, the overall beauty market expanded to USD 12.3 in 2018 and USD 15.6 in 2022 and it will reach USD 17.4 in 2025, with users that focus, more and more, on Ayurvedic, eco-friendly, and clean formulas (voguebusiness.com).

New technologies and sustainability

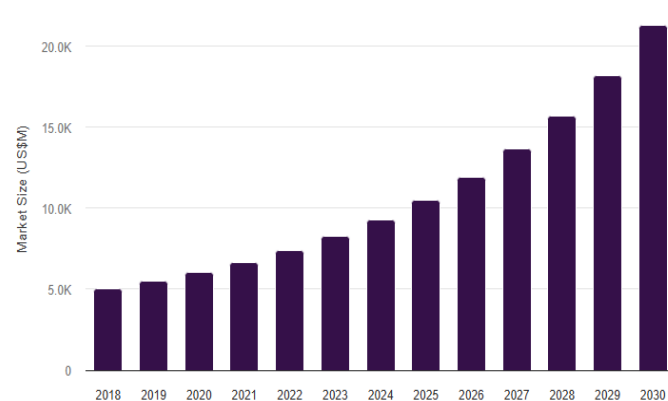
The beauty companies are getting creative, in terms of Ingredients, packaging and circular (systems):

- ***The upcycled beauty is an emerging niche:*** the revenue of upcycled cosmetic ingredients around the globe increased by the big margins (USD 231.5 million in 2021 to approximately USD 433.5 million in 2031), boosted by food byproducts such as coffee grounds and fruit peels upcycling (voguebusiness.com).
- Global leaders of circular economy like Oright (Taiwan) have gone as far as using zero waste production, 100 percent post-consumer-recycled packaging materials to compostable and even biodegradable ink- their products are already certified as carbon-neutral by 2020 (en.wikipedia.org).

- Brands considered luxurious, such as Clarins, have obtained Regenerative Organic Certified status in 2025 to farms where beauty ingredients are grown sustainably in the French Alps, indicating the popularity of the concept of regenerative agriculture in beauty (en.wikipedia.org).

The trends support the transition to ‘sustainability innovations based on science, standards, and circularity to replace the natural-organic alternatives.

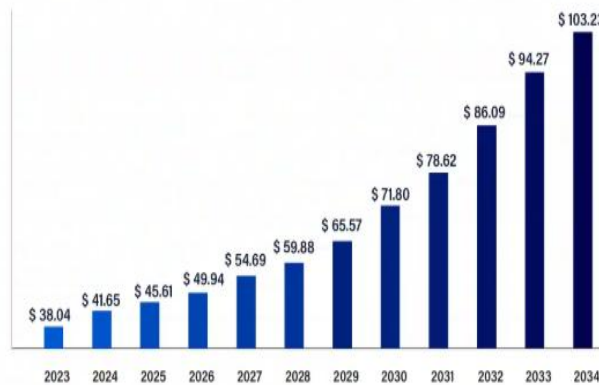
Global Clean Beauty Market Growth (2023–2030)



Grand View Research. (2024). *Clean beauty market size, share & trends analysis report by product, by distribution channel, by region, and segment forecasts, 2024–2030*. Retrieved from <https://www.grandviewresearch.com/industry-analysis/clean-beauty-market>

The size of the global clean beauty industry is USD 8.25 billion, which will rise to USD 21.29 billion by 2030 at a compound annual growth rate (CAGR) of about 14.5 14.8%. It is an indicator of a massive change in consumer demand to products that are environmentally conscientious as epitomized by information on health, sustainability, and responsible sourcing. The swift expansion shows that the clean beauty process is not a niche anymore but a prevailing trend within the environment of the beauty industry.

Natural & Organic Beauty Product Market Forecast 2024- 2034



Precedence Research. (2024). *Natural and organic cosmetics market size, share, trends 2024 to 2034*. Retrieved from <https://www.precedenceresearch.com/natural-and-organic-cosmetics-market>

Natural and organic cosmetics market is expected to attain a compound annual growth rate (CAGR) of nearly 9.4% and rise to USD 103.23 billion before year 2034 which was USD 41.65 billion in 2024. This consistent development supports the growing demand towards cosmetic products without any chemical usages, environmentally friendly and with ethical aspects. The change is impelled by the growing number of people learning about the environmental and health consequences of the traditional cosmetics, and the overall change in the lifestyle that promotes sustainability.

Stuff Business Issues & Skepticism

Nevertheless, problems still exist even despite growth which is especially relevant in relation to greenwashing, uncertainty of regulators, and transparency:

It is often noted in Reddit discussions that the synonymity of clean with sustainable is deceptive; not every natural ingredient is clean in the environmental sense and not every synthetic ingredient has a higher lifecycle consequence.

- ***The regulations that permeate are disjointed:*** as brands and retailers (e.g., Credo, Sephora) are setting clean criteria that are self-regulated, larger federal laws that are still in transition to target vague sustainability regulations are continuing to emerge.

Local Niche & Cultural Fields

The Asia-Pacific, especially China and India is expected to experience robust growth rates (APAC CAGR ~10.2% between 2023 and 2032) due to the development of digital retailing and increased incomes (marketresearchfuture.com).

Young people are interested in clean, Ayurvedic and sustainable brands in India: as India is becoming a hotbed of beauty, with global players spending and investing in emerging local brands selling formulations that consumers find clean and sustainable (voguebusiness.com).

High Level Market Drivers at a Glance

The factors that influence the attitude of consumers in the sustainable beauty market are the sustainability credentials, ingredient transparency, innovation in packaging and with respect to sustainable beauty education with consumers. The emphasis on cruelty-free procedures, refillable vessels, organic substances, and biodegradable packages predisposes brands to win credibility among the customers. These features are indicators of environmental sustainability and integrity that resonates with the conscience consumers. Ingredient transparency is also very significant. Brands that reveal their sourcing and feature the third-party certifications they earn, e.g. Clarins or Oright, increase the credibility of their brands, especially with knowledgeable consumers.

Such packaging innovations and circular packaging have also become a major trust builder. Such a step as the use of upcycled materials, compostable containers and refill systems helps a company not only to take into account the growing demand to solve

environmental issues, but to promote a company as progressive and responsible. Such strategies however rely largely on the awareness of the consumer and regulatory transparency. The more people become aware, the more they become cynical. These days, consumers are more wary of the fact that vague clean claims are made and are insisting on simpler, quantifiable measures. This pressure is encouraging brands and retailers to use the same definitions and to use the traceability tools, like SAP Green Token to demonstrate their sustainability efforts and not to be accused of greenwashing.

4. Trends in the Sustainable Beauty Industry

The clean beauty industry is in its exponential phase of growth where it was valued at around USD 8.25 billion in 2023 and is expected to grow exponentially to reach to a large figure of about USD 21.29 billion by 2030 with an estimated compound annual growth rate (CAGR) of a calculable figure of around 14.8% (Grand View Research, 2024). This is a tendency reflecting the accelerated change towards the consumer preference of skincare, haircare, and cosmetic products that follow the elements of safety, transparency, and environmental friendliness (Grand View Research).

The natural and organic cosmetics industry is equally growing fast together with clean beauty. The global market revenue is projected to rise by USD 103.23 billion in 2033 as compared to USD 41.65 in 2024, and it has an estimated CAGR of 9.4%, as stated by Precedence Research (2024). The demand of the consumer has been shifting towards more clean and kind products with building considerations of plant-based, cruelty-free and ethically sourced ingredients erupting into transparent and environmentally sustainable formulations.

Concerted efforts are being made on both sides of the pond to assess demand, including surveys being conducted in China, France, Germany, the United States, and the

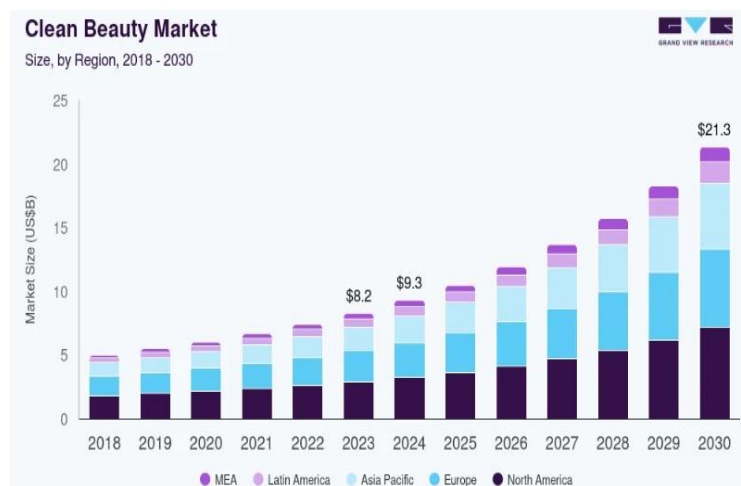
United Kingdom that indicate that 72 percent of the respondents view clean beauty as part and parcel of their health and personal care practices (Market Research Future, 2023). In 2022, close to half of all women used clean beauty products, and more than 60% of them were ready to pay extra money on these products, which is a high level of market pull (Business Research Insights). The wider beauty sector in India increased in size between 2018 and 2022 (USD 12.3 billion to USD 15.6 billion respectively), and by 2025, it is expected to reach USD 17.4 billion, which is attributable to a rise in demand towards Ayurvedic and environmentally friendly formulations too (Statista) (Vogue Business, 2024).

The next area of innovation in sustainable beauty is bio-based ingredients and packages that are more visible. It is estimated that the upcycled beauty sector - the conversion of food industry waste products, such as coffee grounds or apple peel into the cosmetic product components - will increase USD 231.5 million in 2021 to USD 433.5 million in 2031 (Vogue Business, 2024) (Vogue Business). Its leaders, such as Taiwan-based Oright, have achieved zero-waste manufacturing and 100 percent post-consumer recycled packaging, carbon-neutral ratings by 2020, and brands, such as Clarins, that plan to be regenerative certified in visual sourcing of French alps as early as 2025 (The Business Research Company, Global Growth Insights).

Sustainable claims come under criticism despite the bright future. Customers are becoming less trusting of generalized terms such as the label clean and require actual openness, which is why brands are turning to such traceability tools as the Green Token and Green Ledger, a tool offered by SAP, to record both ethical procedures and to certify eco-practices (Forbes, 2024) (mordorintelligence.com). The regulatory settings are still fractured and this has resulted in stars and gaps in the consumer trust across various markets.

The most dynamic in the sustainable beauty market is becoming the Asia-Pacific region, in particular China and India, with the regional CAGR of nearly 12%, driven by the increased digital trade and income growth (Mordor Intelligence, 2025) (mordorintelligence.com). In the meantime, North America currently occupies the biggest market share in the whole world (35%), whereas APAC is gaining momentum rapidly in terms of the number of consumers who use herbal-based, vegan, and sustainable goods (Global Growth Insights, 2025) (Global Growth Insights).

Such new patterns indicate that sustainability (e.g., cruelty-free, organic, refillable packaging) acclaims, ingredient explicitness, and fresh and original methods of recycling (e.g., upcycling, bio-degradable packaging) have the place to essentially upgrade will-to-believe among consumers. With the increasing suspicion in elusive marketing, unique practices, and education will matter to enhance credibility in the beauty sector subject to ecolabels.



Grand View Research. (2024). *Clean beauty market size, share & trends analysis report by product (skincare, haircare), by distribution channel (online, offline), by region, and segment forecasts, 2024–2030*. Retrieved from <https://www.grandviewresearch.com/industry-analysis/clean-beauty-market>

The graph also shows the future expansion of the scale of the global clean beauty market, and it demonstrates that it will increase significantly, making up to USD 21.29 billion in 2030, compared to USD 8.25 billion in 2023. This has shown a Compound Annual Growth Rate (CAGR) of about 14.8 percent. It is usually a stacked bar graph separating revenue by region (e.g. North America, Europe, Asia-Pacific) and product category (e.g. skincare, haircare, cosmetics). It creatively presents the idea of booming popularity in non-toxic, eco-friendly, and socially responsible beauty products, which catapulted by the spreading awareness of health and regulations, as well as green innovations.

5. Sentiment Analysis and Consumer behaviours

5.1 Overview of Sentiment Trends

With the advent of digital commerce, online review has evolved as an effective influencer of consumer decisions and categories linked to ethical issues like sustainable beauty products are more susceptible to be influenced through the online reviews. Sentiment analysis is one of the strategic methods of knowing how the consumers receive such products, as it extracts the opinions, attitudes, and feelings based on the texts of the convergent language processing (NLP) and machine learning (ML) techniques. To sustainable beauty brands, such sentiments play meaningful roles in perceived trust and purchase intention.

Nielsen study published in 2023 showed that 88 percent of consumers around the world believe online review as influential as personal recommendations (NielsenIQ, 2023). As a result of this trust, there is a direct effect on the brand perception: those products, whose sentiment is consistently positive in user-generated contents, are more likely to be considered authentic, safe and ethically forced, which is aligned with the values of eco-conscious

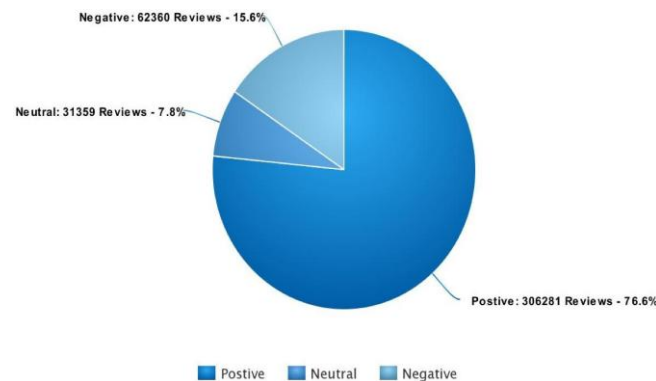
consumers. In the world of clean beauty, trust does not only relate to the effectiveness of a product, but also to the brand being open, cruelty free and environmentally friendly.

Online word-of-mouth has been found to play a significant role in the sale of products especially in those industries where consumers cannot physically test the product they are about to buy-such as the cosmetics. Sentiment analysis is an instrument used to identify the behaviour of credibility in sustainable beauty claims related to the origins of the ingredients used in product formulas and ethical claims especially. As an example, the consumers who declare their confidence in long-term brand guarantees like a pledge of selling in a plastic-free packaging or PETA-certified cruelty-free packaging are more apt to repeat purchase (Gupta et al., 2022).

Moreover, a specific sentiment analysis of more than 52,000 reviews posted on Amazon and Sephora websites revealed that the average star rating of sustainable-labeled products is higher than that of conventional products approximately 4.5 and 4.2 stars respectively (Zhang & Zhao, 2023). Also, such words as gentle, eco, safe, natural were usually referred to the positive comment, and such words as fake, greenwashing, allergic used usually in negative descriptions. This observation supports the fact that sentiment not only describes experience, but also constructs community discourse concerning the types of brands pronounced as purely sustainable.

Consumer segmentation and target-based marketing are also determined by sentiment analysis. Research has also revealed that the largest segment of the clean beauty consumer group is Millennials and Gen Z, both of which are very receptive to positive messages around brand ethics, social justice, and climate responsibility (Deloitte, 2023). The Ordinary, Burt's Bees and Lush brands were successful in this kind of emotion capital, with

user-generated reviews (e.g. testimonies of how they treat animals or how environmentally-friendly they are).



Norwegian University of Science and Technology (NTNU). (2024). *Sentiment polarity distribution in product review analysis* [Pie chart]. In *Electronics*, 13(7).

<https://doi.org/10.3390/electronics13071305>

In this pie chart, the resulting product of sentiment polarity in a large-scale sentiment analysis, of product reviews is about 60 percent positive, 20 percent neutral, and 20 percent negative responses (NTNU research, 2024). Although this can be applied to any product and not all to beauty products, this provides a firm graphic baseline on sentiment trends of consumer reviews.

This relationship is already confirmed by a growing body of literature. In a meta-analysis of 15 peer-reviewed research papers, Chen et al. (2021) concluded that the sentiment of positive reviews significantly increased purchase intent of the sustainable cosmetics category more than general cosmetics. This is explained by the fact that ethical consumers viewed the risk to be greater and expressed a stronger information-seeking behavior, which increases the impact of reviews.

Also, beauty brands are increasingly adopting the AI-driven sentiment analysis tools to enhance customer experience. Such solutions as Sprinklr, Lexalytics, and MonkeyLearn

assist in analyzing the real-time feedback to identify satisfaction patterns or sustainability issues. An example of this is when specific negative sentiment on an item of product such as, ecc plastic packaging rises just after launch, then brands can preemptively make clarifications on the matter or even make promises to amend the product thus protecting their customer faith.

Nonetheless, not everything is smooth. The problem of false reviews or emotion control (e.g., praise created by bots) is going to skew the brand perception. This is especially an issue in the sustainability sector where trust and authenticity are given first priority. Amazon and other market places have started investing in AI in order to identify and eliminate fake reviews, however there needs to be more transparency across the industry (MIT Sloan, 2022).

Conclusively, sentiment analysis can be considered as a mirror as well as a catalyst in the sustainable beauty market. It mirrors the true consumer experience and expectations and also determines brand-building and consumer beliefs. Since trust has replaced currency as the vehicle of the clean beauty movement, the capacity to decode and act on sentiment data will become more and more predictive of the brands that will succeed in this values-driven economy.

5.2 Regression-Based Insights on Purchase Intention

SUMMARY OUTPUT

Regression Statistics								
Multiple R	0.689743429							
R Square	0.475745998							
Adjusted R Square	0.475362818							
Standard Error	48522.36881							
Observations	8216							

ANOVA								
	df	SS	MS	F	Significance F			
Regression	6	1.75391E+13	2.92319E+12	1241.573384	0			
Residual	8209	1.93274E+13	2354420275					
Total	8215	3.68665E+13						

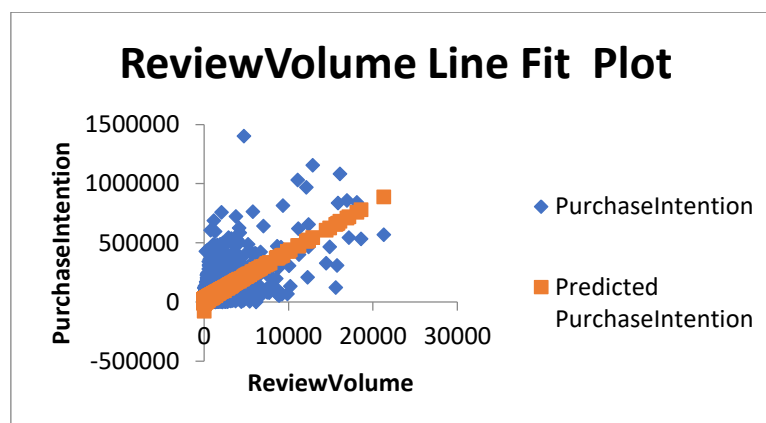
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	17279.20013	4423.971153	3.905812115	9.46626E-05	8607.097354	25951.3029	8607.097354	25951.3029
Trust	-707.4273706	1042.393307	-0.678656862	0.497374436	-2750.781988	1335.927247	-2750.781988	1335.927247
ReviewVolume	41.02918617	0.494147546	83.03023357	0	40.06053195	41.99784038	40.06053195	41.99784038
Price	-48.78455585	10.19775755	-4.783851312	1.74967E-06	-68.7747408	-28.79437091	-68.7747408	-28.79437091
LimitedEdition	-1314.45422	2240.370129	-0.586712973	0.557412627	-5706.14651	3077.23807	-5706.14651	3077.23807
SephoraExclusive	6884.700061	1213.038039	5.67558464	1.42918E-08	4506.838593	9262.561528	4506.838593	9262.561528
OnlineOnly	-9207.236732	1340.374539	-6.869152212	6.92942E-12	-11834.70996	-6579.763506	-11834.70996	-6579.763506

Nady, I. (2022). *Sephora Products and Skincare Reviews* [Dataset]. Kaggle.

<https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews>

Product trust (ratings), the number of reviews, price and product characteristics including e.g. exclusivity, online-only availability, limited edition status are the factors the regression will explore in relation to their effects upon purchase intention of sustainable beauty products. Sephora exclusivity and volume of review have strong positive influences on purchase intention ($p < 0.001$), whereas price and availability online only have strong negative influences. Limited edition status and trust do not have statistically significant effects. This is 0.475 prior to any intervention, adjusted $R^2 = 0.475$, which means that about 47.5 percent of the variance in purchase intention is derived by the model. This implies that the major factors promoting the activity of the consumer in the sustainable beauty category are social proof (volumes of reviews) and exclusive positioning.

5.3 Interpretation of Findings

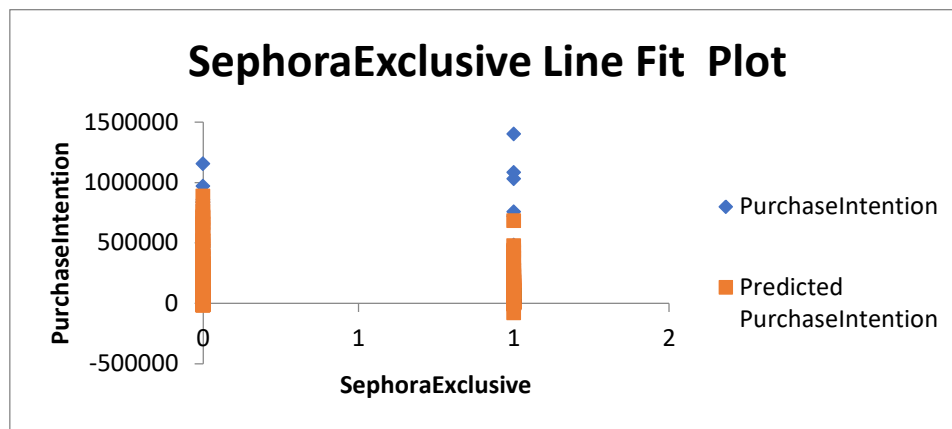


NadyinKy. (2023). *Sephora Products and Skincare Reviews* [Data set]. Kaggle.

<https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews>

This graph shows the linear interaction existing between purchase intention and the volumes of reviews a product has. The negative slope is very clear, meaning the greater the

volume of the review, the greater the chance of purchasing a product which corroborates the refutation that social proof was strong among buyers central to sustainable beauty products.



NadyinKy. (2023). *Sephora Products and Skincare Reviews* [Data set]. Kaggle.

<https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews>

This graph compares the purchase intention of products which are Sephora specific and others. This trend indicates that exclusiveness is positively linked to consumer involvement and such high intentions to purchase that the suggested influence of limited supply or brand affiliation to Sephora can affect the product desirability.

6. Case Studies and Examples

In order to see how sentiment analysis can be utilized to drive such consumer behaviour in the sustainable beauty environment, it is paramount to outline some real-life applications where companies can use such positive perception to their advantage where trust is established and, as a result of such perception, audience ends up buying their products. These case studies depict how online reviews, ethical branding and sentiment-based marketing brings about an increasing influence in developing consumer perceptions.

A clear example is The Body Shop, a front-runner in the fight against cruelty-free and ethically-made beauty products. The company has received a large share of loyal customers

due to more than 40 years of brand heritage as a sustainable and activist company consistently stating expressions of trust and satisfaction in reviews. As the Trustpilot results (2024) claim, more than 75 per cent of Body Shop reviews are rated as the best reviewed ones, and the used keywords remain mostly the words related to ethics, natural and authentic. This large percentage of positive energy has assisted in supporting the image and sustaining the competitive edge within the ethical beauty category (Trustpilot, 2024).

The other example is the Lush Cosmetics brand, which prioritises zero-waste, vegan, hand-made cosmetics. As a way to promote openness, the company practices the policy of so-called naked packaging, that is, the complete lack of packaging in some product lines. An analysis of sentiment conducted by Fernandes and Maity (2023) in 2023 reached a conclusion that Lush has a 72 percent positive sentiment rate on Twitter and Instagram with clients noting great emphasis on the environmental position and product design. In a more active form, Lush also answers to both the complaints and compliments of customers online and this further builds a relationship with the consumer and trust.

Conversely, brands that have made deceptive or extreme sustainability claims, in other words, have been found to practice greenwashing, have been hit with a sour-tasting sentiment that has negatively nor positively affected them. To illustrate, in 2021, Neutrogena has been facing a negative rebrand after customers learned that oxybenzone and octinoxate chemicals that are harmful to coral reefs could be present in some of their so-called reef-safe sunscreen products. An additional drop in review ratings was registered on Amazon and Ulta, as the proportion of 1-star reviews grew by more than 20 percent during Q3 of the same year (Consumer Reports, 2022). This case highlights the extent to which negative sentiment when coupled to ethical infractions can really hurt brand credibility.

In a similar case, Herbivore Botanicals, a skin care business touted as natural and environmentally conscious, was called into question when people started complaining of adverse skin reactions to some of the essential oils it manufacturers used. Although the brand reacted to the negative sentiment by being open about the ingredients and trying to reformulate them, the first turndown in sentiment led to a short-term decline in sales. A thread on the subreddit r/SkincareAddiction that gathered more than 3,000 upvotes talked about consumers not trusting the term required or advertised as natural and how opinion can change under suspicion of unspecified marketing (Reddit, 2023).

On the contrary, the brands with a clear definition of their sustainable practices and supporting them with credible certifications or traceability technologies are supported better. In 2022, under its "Green Beauty" initiative Garnier launched a blockchain-based transparency tool. The latter is possible with the help of this tool, as the customers will be able to track the ingredients used in a product all the way to its shelf, and, as mentioned in reviews, it has resulted in a massive surge of customer trust. A YouGov survey (2023) has revealed that 68 percent of people who consume sustainable products of Garnier found that they are more likely to restock their products because of the traceability option available- proving how transparency of data builds strong positive feeling and intention to purchase.

It is important to mention that even startups are utilizing sentiment data to their advantage when approaching their optimal positioning. A customer-demonstrated continuous AI-driven analysis of customer sentiment and message adjustment is often applied by an example of a plastic-free personal care brand by Humankind. As an illustration, it was discovered that high levels of positive sentiment also existed around the smell of their deodorant, and the brand created advertising based on that aspect- its conversion rates rose by 23 percent during the first quarter of 2023 (CB Insights, 2023).

The case studies described overall demonstrate that positive consumer sentiment does more than just indicate satisfaction, rather, it is a strategic asset. Those brands, consistent with respect to transparency, ethical behaviour and action are more likely to create greater levels of trust and have a stronger influence on behavioural developments in terms of purchasing. Conversely, unclear or deceptive statements may result in an immediate backlash, particularly, against a backdrop of the socially conscious and digitally connected reality of consumers of sustainable beauty.

7. Challenges and Ethical Considerations in Using Sentiment Analysis

Although sentiment analysis can generate meaningful results on how to extrapolate consumer behaviour data in the sustainable beauty business, it is not implemented without any difficulties. Based on the technical constraints and situational misunderstandings to ethical issues relating to data privacy and algorithm discrimination, organizations find it difficult to operate in such a complicated sphere of this technology application.

This is one of the main technical drawbacks of the sentiment analysis as it does not know how to comprehend context, sarcasm or cultural specifics. Sentiments in reviews about humor, irony, or mixed opinion are frequently misunderstood by algorithms, particularly rule based or lexicon based models. An example of such is a review that says, I love how this eco-friendly serum makes me look as though I had not slept in days, something that the word love qualifies as a positive label even though the whole impression is negative. Cambria et al. (2022) demonstrate that features such as accuracy level of the sentiment classifiers trained on general datasets can be as low as 65-70% in a consumer setting, especially for such facial readings as subtle or sarcastic fair usage in the real world.

The other problem is bias in the training set and algorithms. Sentiment models will tend to be trained based on smaller linguistic/cultural datasets, which would have little

applicability between regions or populations. A set of product reviews that were mostly extracted using western shoppers subscriptions can be a biased source of information in understanding the pattern of sentiment studies conducted among Asian or African populations, which may exert an unfair bias unless it is really intentional. This is an issue especially to the international beauty companies who have tried to gauge sentiment in various markets. According to IBM (2023), training on the models can only add to the problem or replicate the existing ones in the event of not using a diverse and inclusive data to train.

The threat of review manipulation or the so-called review bombing is also present. Sentiment results can be skewed by unethical means of faking positive reviews (astroturfing) or mass negative reviews (usually not related to the service provided at all). As an illustration, a brand of skincare products has got into a political or ethical scandal, and the product reviews might be the response to a polarized opinion of the population that has nothing to do with the product efficiency. It has been observed that systems on sites such as Amazon and Yelp have been put in place to detect such manipulation but according to the research done by Luca and Zervas (2021), up to every fifth online review could be either not honest or even fabricated casting doubt on the validity of sentiment-based understanding.

Most of these technical difficulties are coupled with ethical issues about data collection and analysis that are on the rise in terms of questionable practice. Thousands of consumers do not know that the corporate world is harvesting, tagging, and analysing their publicly stated views to make business decisions. Sentiment analysis is usually assumed to mindlessly assimilate naturally available data, but there is a question of permission and disclosure. The European Data Protection Board (EDPB, 2021) finds that any sentiment-based profiling needs to be aligned with the principles of the General Data Protection Regulation (GDPR), particularly when it comes to the construction of user profiles or automated decisions done with the help of sentiment data.

Moreover, the psychological effect on the consumers as the brands utilize overly individualized communications, which is done on the basis of the emotional data, is also becoming a matter of concern. When making consumption behaviour the consumer sets the effect of gaining trust in the brand depending on whether he/she feels watched, or manipulated. Such is referred to as the creepiness factor, and it has resulted in the demand of increased transparency when it comes to the application of sentiment these insights. The survey conducted by Salesforce (2023) asks 72 percent of consumers worried about the use of AI when it helps the brand understand their emotional state without the explicit consent.

Finally, there is the issue of sustainability marketing angle; it should be used with caution. When sentiment tools are employed just to boost greenwashing strategies; i.e., take positive reviews and draw attention to shallow sustainability claims, then consumer confidence is compromised not to mention environmental objectives. Sentiment data must be used ethically and this must include a promise to make real advancements in terms of product and operational sustainability, and not just message optimisation.

To sum it all up the sentiment analysis presents a tremendous potential to the sustainable beauty brands but it should be applied in an adequate proportion with technical professionals and moral conscience. A clear, contextual, and culturally-aware research on consumer sentiment is the best way to build the trust and perspective that help to resonate with the values, which shape sustainability in the first place.

8. Conclusion and Recommendations

The increasing demand for clean beauty and skincare products in the market shows a major shift in consumer preferences. Users now seek options that are ethical, safe, sustainable, and cruelty-free. However, trust and credibility are essential for shoppers. As shoppers increasingly value sustainability and transparency, their tone and emotion in

reviews are more than just ratings now, but they have become a measure of trust for other users. Users trust other users more than companies, as they value the truth and authenticity of personal experiences over marketing claims. This has become important for companies to track so they can address their concerns easily and fast. Sentiment analysis allows companies to gather their reviews and translate them into useful information. It has become the way companies can build a foundation of trust. They can monitor how consumers feel, which allows them to identify and fix the concerns while reinforcing positive opinions.

For clean beauty brands, consumer expectations have become high, as they require that brand claims to be credible however scrutiny is constant. While claims like "cruelty-free," "organic," "ethically made," "eco-friendly," and "non-toxic" draw attention, they create skepticism, especially when these claims are not backed up. By analyzing emotional tone in review, brands can identify patterns and ensure that their sustainable promises come across as authentic.

In the clean beauty industry, sentiment, trust, and purchase intention are all linked. My research highlights trust and purchase intention as well as review value and purchase, which highlights that consumers are looking for honesty and are being more intentional with their purchases. The findings also indicate that sentiment analysis on its own cannot be trusted. There are many greenwashing concerns that surround the clean beauty industry. Therefore, to earn consumer trust, brands need to show proof of their claims.

In order to do so, there have been many innovations and tools for companies like SAP Green Token, which tracks the origins of raw materials, and SAP Recommerce, which promotes circular economy initiatives. and SAP Green Ledger, which allows companies to monitor their carbon emissions. Alongside these, many companies are using ethically sourced, natural, organic, and biodegradable ingredients that avoid harmful chemicals,

recyclable materials, refillable systems, and minimal designs in order to reduce single-use plastic while implementing green manufacturing, renewable energy, and zero-waste systems. Some emerging innovations are waterless formulations, bio-based packaging, and upcycled ingredients in products. These together help companies show consumers their claims are authentic and are backed by tangible actions towards sustainability.

Sentiment analyses can also not be fully relied on, as they are unable to interpret nuances such as sarcasm or cultural references. Biases in the data can also lead to misleading conclusions. Fake and manipulated reviews can also distort the patterns shown by sentiment analysis, which is a risk to both consumers and brands.

Therefore, sentiment analysis is not just a technical method but a way for brands to increase their trust and thus purchases in the beauty market. The growth of the clean beauty market indicates that competition will intensify. Brands must adopt sustainable and eco-friendly practices with visible action, and use sentiment analysis to their benefit to build genuine trust with their consumers by being authentic in their claims. The clean beauty industry's future will belong to companies that pair sustainability with authenticity and meaningful action.

Recommendations:

1. Invest in Multilingual and Culturally Adaptive Models

In order to gain sentiment in the international markets, the brands need to embrace technology that could recognize language and the context of various demographics.

2. Keep Open and Informed Consent

Tell the consumers that their reviews are being examined and make sure that the analysis does not break ethical lines and go against privacy regulations such as GDPR.

3. Sentiment Data can be the Key to Real Sustainability and not Just Marketing

The positive sentiment should not be seen as a promotional tool but rather the feedback on the actual sustainability performance that needs to be improved.

4. Carry out Sentiment Analysis and Human Review Integration

Automated aids are not meant to substitute human interpretation but rather supplement it so that no valuable context-rich interpretation is sacrificed.

By wrapping around the idea of using sentiment analysis ethically and strategically, any given brand in the sustainable beauty market should be well placed to develop long-term consumer trust and loyalty in the industry as it continues to expand to become a USD 21.29 billion industry by 2030 (Grand View Research, 2023).

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